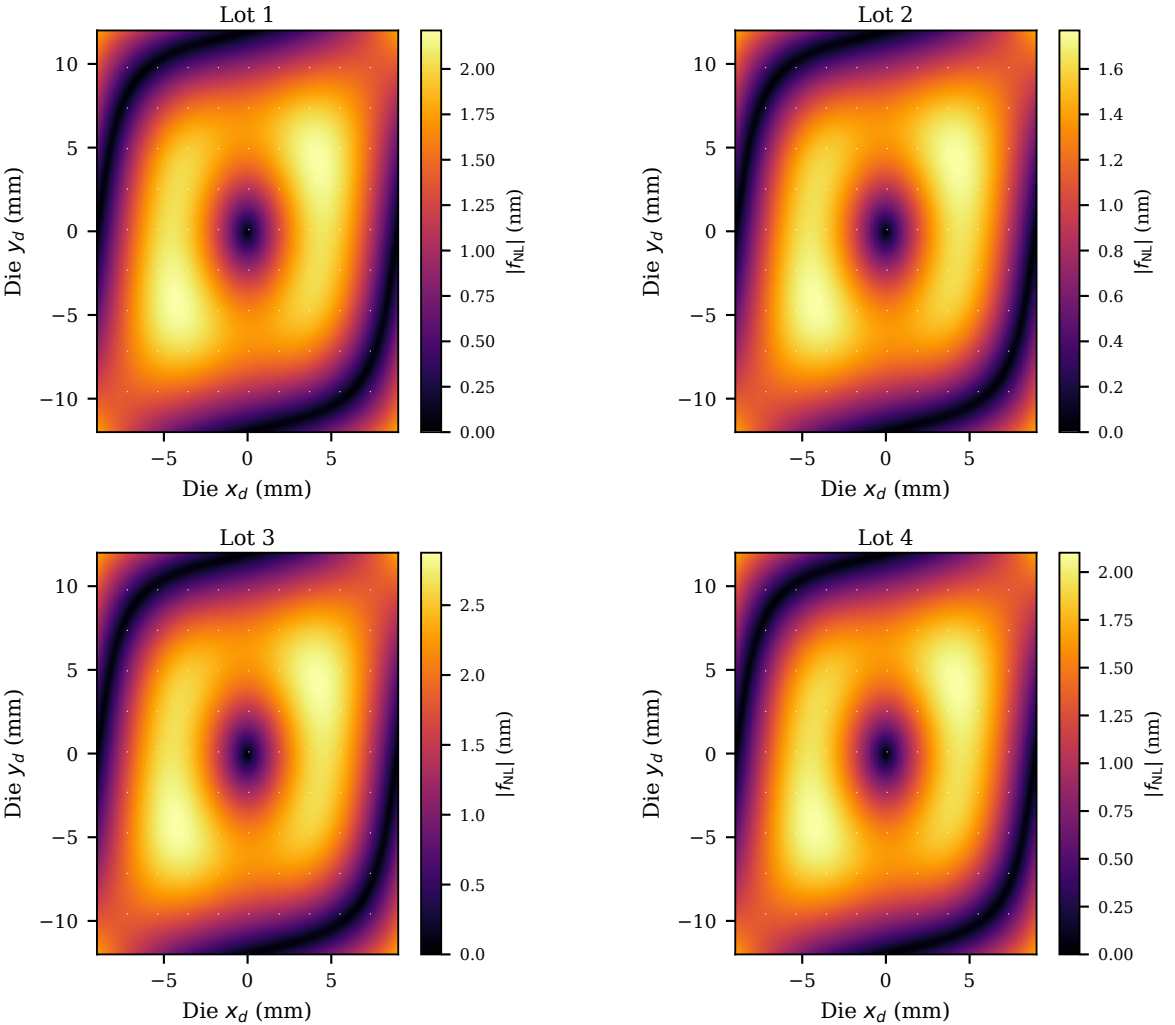Nonlinear intra-field residual spatial pattern (mid-lot wafer)